

Activity Diagram

The diagram models the NBP Analytics application across three swimlanes: User, App/UI, and NBP API. After opening the app, the user selects one of two pages. The Session Trends branch fetches historical exchange rates and renders a line chart with statistics. The Histogram branch fetches one or two currency series in parallel, computes daily percentage changes, and renders a distribution bar chart.

Application initialization and page selection

The first part of the diagram presents the startup sequence of the NBP Analytics application. When the user opens it in a browser, the application mounts its root component, sets the default view to Session Trends, and renders a navigation dropdown. The user then selects a page, which triggers a conditional branch: the flow proceeds either to the Session Trends panel or to the Change Distribution Histogram panel. This decision point serves as the entry gate for the two main modules of the application.

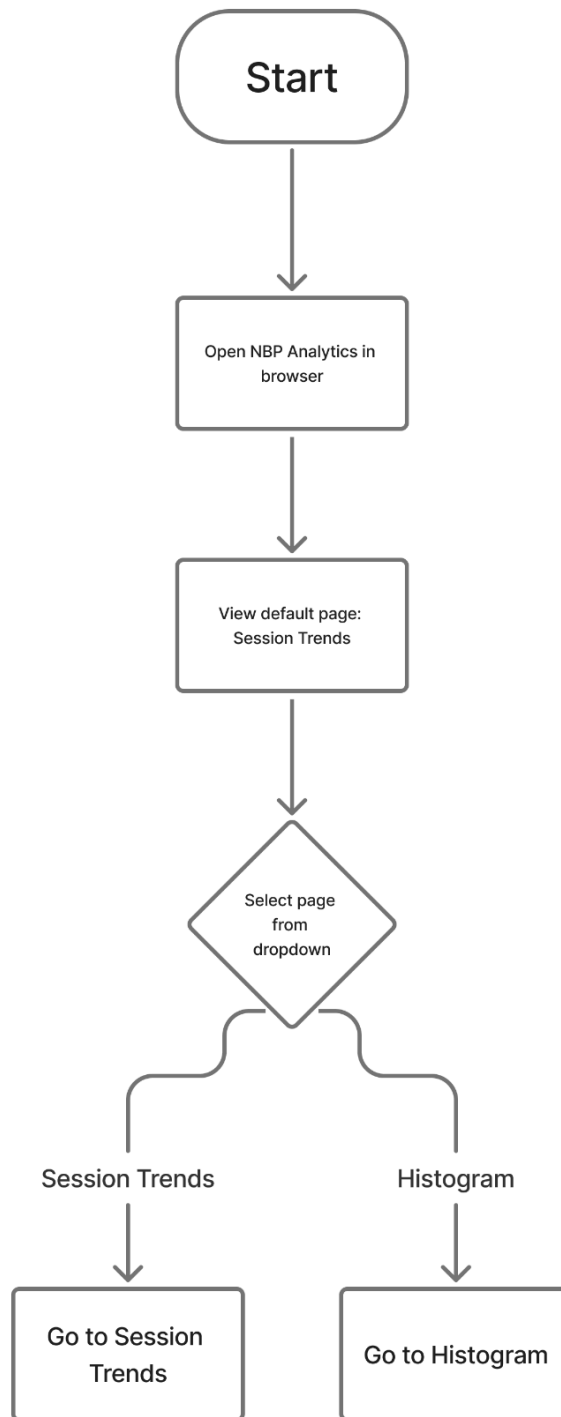
Session Trends panel workflow

The second part describes the repeating workflow of the Session Trends panel. On load, the application initializes its state with default values: currency set to USD, start date to January 2, 2002, and period to one month. The end date is computed automatically. In each loop iteration, the application sends a GET request to the NBP API for the selected currency and date range. On success, it parses the response, calculates statistical measures (min, max, average, median, mode, standard deviation, and coefficient of variation), and renders a line chart with accompanying stat badges. On failure, an error message is displayed. The user may then change the currency, start date, or period - each triggering a new iteration - or navigate to another page, which ends the loop.

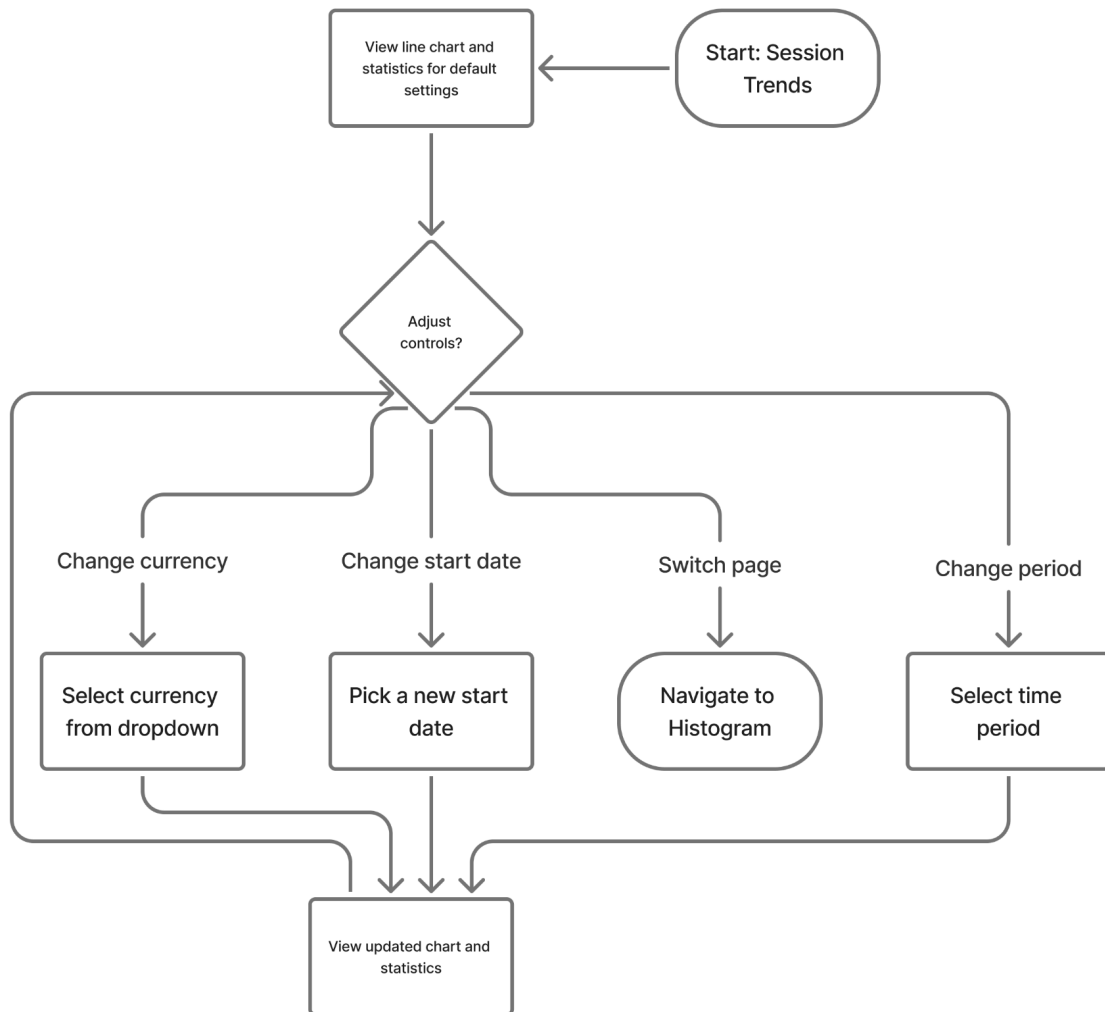
Change Distribution Histogram panel workflow

The third part illustrates the workflow of the Histogram panel. The application initializes with two currencies (USD and EUR), quarterly mode, and the most recent quarter selected. Each iteration resolves the date range and checks whether both currencies are the same - if so, a single API request is made; otherwise, two requests are sent in parallel. On success, the application calculates daily percentage changes and groups them into histogram bins from -3% to +3% in 0.5% steps, rendering the result as a bar chart with hover tooltips. On failure, an error message is shown. The user may adjust either currency, switch between monthly and quarterly mode, or select a different period - any change restarts the loop. Navigating to another page ends it.

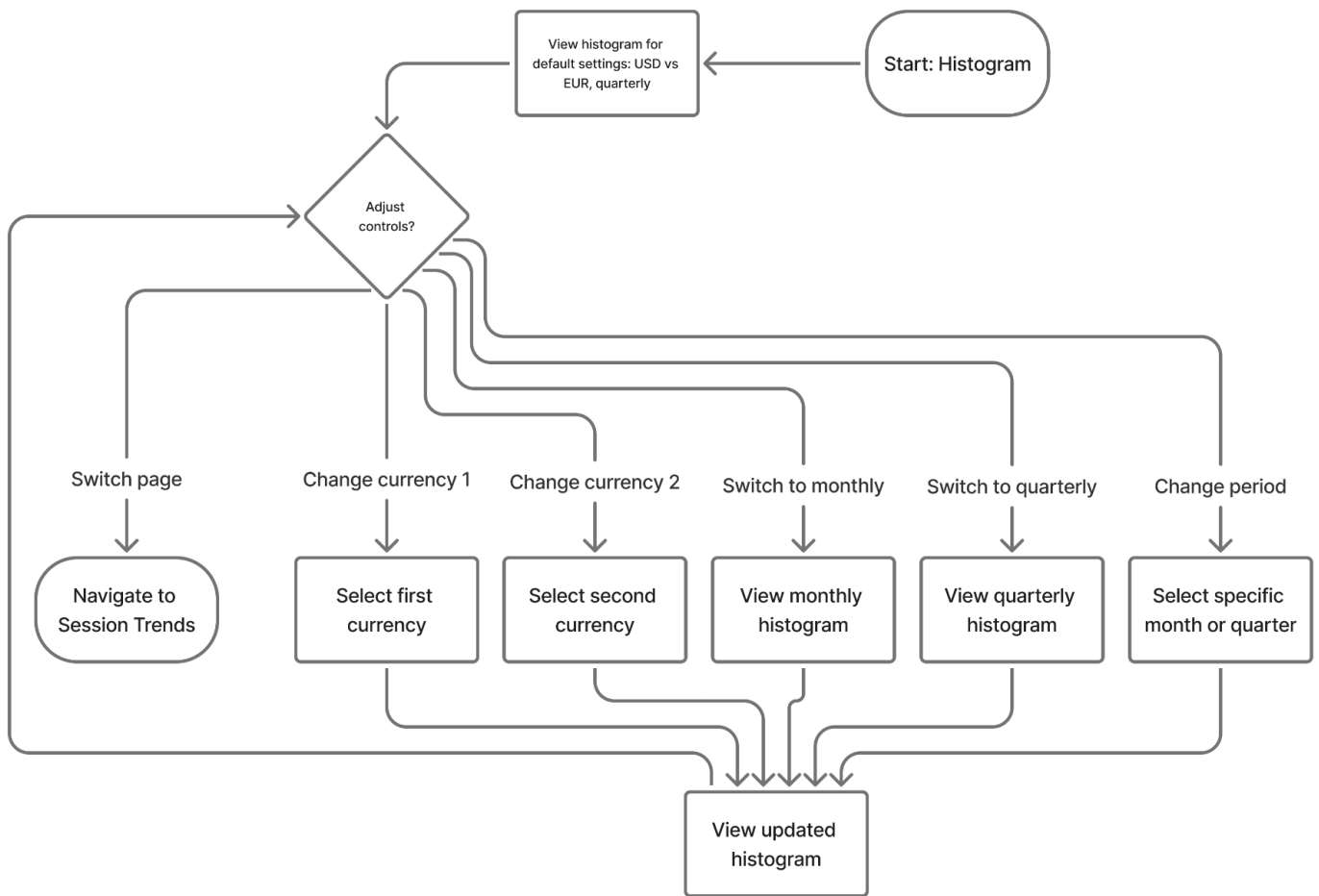
Application initialization and page selection



Session trends panel workflow



Change Distribution Histogram Panel workflow



Error handling

